

1 Short note on research plan

Setting up the model: There are two ages, young and old, respectively denoted by $a \in \{y, o\}$. Furthermore, there are two types of job, high and low, receptively denoted by $j \in \{h, \ell\}$. Time is denoted t . The labor supply is denoted by l , and for young workers it is

$$l_t^y = l_{h,t}^y + l_{\ell,t}^y = N_t^y$$

where N_t^y is an exogenous young cohort size. The labor supply of older workers is equal to the previous period's supply of younger workers, adjusted by job-specific retention rates

$$l_t^o = l_{h,t}^o + l_{\ell,t}^o, \quad l_{h,t}^o = \rho_h l_{h,t-1}^y, \quad l_{\ell,t}^o = \rho_\ell l_{\ell,t-1}^y,$$

where ρ_h and ρ_ℓ are the retention rates of older workers. I study an increase in the number of workers who were employed in high-level jobs in the previous period. Formally, the comparative static is taken with respect to $l_{h,t-1}^y$, holding $l_{\ell,t-1}^y$ fixed:

$$\frac{\partial l_{\ell,t-1}^y}{\partial l_{h,t-1}^y} = 0.$$

This exercise should therefore be interpreted as an increase in the size of the previous high-job cohort, rather than as a reallocation of a fixed previous young cohort from low-level to high-level jobs. A higher retention rate ρ_h would operate through the same state variable, $l_{h,t}^o$, and therefore generates qualitatively similar effects.

The productivity of young workers is

$$L_t^y = \theta_h^y l_{h,t}^y + \theta_\ell^y l_{\ell,t}^y,$$

where θ is marginal productivity. The productivity of older workers is

$$L_t^o = \theta_h^o l_{h,t}^o + \theta_\ell^o l_{\ell,t}^o.$$

I assume that high-level workers are more productive than low-level workers, hence

$$\theta_h^o > \theta_\ell^o \quad \text{and} \quad \theta_h^y > \theta_\ell^y.$$

For revenue, the firm has the standard Cobb-Douglas production function with a technology aspect, A_t ,

$$\begin{aligned} \text{revenue}_t &= A_t Y_t = A_t (L_t^y)^\alpha (L_t^o)^{1-\alpha} \\ &= A_t \left(\theta_h^y l_{h,t}^y + \theta_\ell^y l_{\ell,t}^y \right)^\alpha \left(\theta_h^o l_{h,t}^o + \theta_\ell^o l_{\ell,t}^o \right)^{1-\alpha} \end{aligned}$$

For completeness, $\frac{\partial Y_t}{\partial L_t^y} > 0$, $\frac{\partial Y_t}{\partial L_t^o} > 0$, $\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} < 0$, $\frac{\partial^2 Y_t}{\partial L_t^o \partial L_t^o} < 0$ and $\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} = \frac{\partial^2 Y_t}{\partial L_t^o \partial L_t^y} > 0$. This is similar to the Bianchi-Paradisi-paper. The firm chooses the number of young workers in bottom and top jobs. The maximization problem is

$$\max_{l_{h,t}^y, l_{\ell,t}^y} A_t Y_t - \sum_{a=y,o} \sum_{j=h,\ell} w_{j,t}^a l_{j,t}^a - \frac{c}{2} K_t^2,$$

where $K_t = l_{h,t}^o + l_{h,t}^y$. The fixed young supply condition $l_{h,t}^y + l_{\ell,t}^y = N_t^y$ is imposed in equilibrium. Hence, the FOCs below should be interpreted as inverse labor-demand conditions that are combined with the fixed supply of young workers, such that wages adapt to the quantity of workers. I also assume that the wages of older workers are equal to the previous period's wages:

$$w_{h,t}^o = w_{h,t-1}^y \quad \text{and} \quad w_{\ell,t}^o = w_{\ell,t-1}^y.$$

Finally, I assume

$$w_{h,t}^y = \mu^y w_{\ell,t}^y \quad \text{and} \quad w_{h,t}^o = \mu^o w_{\ell,t}^o.$$

In equilibrium, the following relations hold,

$$\begin{cases} w_{\ell,t}^{y*} &= A_t \theta_\ell^y \frac{\partial Y_t}{\partial L_t^y} \\ w_{h,t}^{y*} &= \mu^y A_t \theta_\ell^y \frac{\partial Y_t}{\partial L_t^y} \\ K_t^* &= \frac{A_t \frac{\partial Y_t}{\partial L_t^y}}{c} (\theta_h^y - \mu^y \theta_\ell^y) \end{cases}$$

Average wages of young are

$$\bar{w}_t^y = \frac{l_{\ell,t}^y}{l_t^y} w_\ell^y + \frac{l_{h,t}^y}{l_t^y} w_h^y.$$

Similar to Bianchi and Paradisi, the average wages of young workers change with the following:

$$\frac{\partial \bar{w}_t^y}{\partial l_{h,t-1}^y} = \underbrace{\frac{1}{l_t^y} (\mu^y - 1) w_\ell^y \frac{\partial l_h^y}{\partial l_{h,t-1}^y}}_{\text{Career spillovers} < 0} + \underbrace{\left[\frac{1}{l_t^y} (\mu^y - 1) l_h^y + 1 \right] \frac{\partial w_\ell^y}{\partial l_{h,t-1}^y}}_{\text{Wage level} > 0}$$

$$\text{when } c > \bar{c} = A_t \underbrace{(\theta_h^y - \mu^y \theta_\ell^y)}_{>0} \underbrace{\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o}}_{>0} \theta_h^o.$$

Solving the model: The total supply of young workers is fixed at N_t^y . As the firm's first-order conditions are interpreted as inverse labor-demand conditions for young workers in high and low jobs, the equilibrium wages adjust such that the firms desired allocation satisfies the fixed young condition,

$$l_{h,t}^y + l_{\ell,t}^y = N_t^y.$$

Using the equilibrium of the high level positions, $l_{h,t}^y$ is found by solving

$$K_t^* = \frac{A_t \alpha (L_t^y)^{\alpha-1} (L_t^o)^{1-\alpha}}{c} (\theta_h^y - \mu^y \theta_\ell^y)$$

$$l_{h,t}^o + l_{h,t}^y = \frac{A_t \alpha [\theta_\ell^y N_t^y + (\theta_h^y - \theta_\ell^y) l_{h,t}^y]^{\alpha-1} (L_t^o)^{1-\alpha}}{c} (\theta_h^y - \mu^y \theta_\ell^y).$$

Here, $l_{h,t}^o$ and L_t^o are fixed from the previous period. Furthermore, I assume A_t is a known process, N_t^y is exogenously given and α , θ_h^y , θ_ℓ^y , μ^y and c are constants to calibrate. While the problem cannot be solved analytically, it is one-dimensional, and I can solve numerically for $l_{h,t}^y$. Once $l_{h,t}^y$ is found, I can find $l_{\ell,t}^y$ as well as wages, $w_{\ell,t}^{y*}$ and $w_{h,t}^{y*}$ from the equilibrium conditions.

Simulating the model: To simulate the model, I need to choose a certain set of parameters. I choose the baseline in Table 1.

Table 1: Baseline Parameter Values

Parameter	Value	Description
A_t	5	Technology/productivity shifter
N_t^y	1.9	Young cohort size
α	0.9	Output elasticity of young effective labor
θ_ℓ^y	0.66	Productivity of young workers in low-level jobs
θ_h^y	5	Productivity of young workers in high-level jobs
θ_ℓ^o	0.9	Productivity of old workers in low-level jobs
θ_h^o	4.7	Productivity of old workers in high-level jobs
μ^y	7.3	Young high-job wage premium
ρ_h	0.5	Retention rate from high-level jobs
ρ_ℓ	0.6	Retention rate from low-level jobs
c	0.5	Cost parameter for high-level positions

Furthermore, I need to choose a starting set of initial base conditions as seen in Table 2.

Table 2: Baseline Initial Conditions

Variable	Value	Description
$l_{h,0}^o$	0.2	Initial old workers in high-level jobs
$l_{\ell,0}^o$	0.45	Initial old workers in low-level jobs
$w_{h,0}^o$	0.5	Initial wages for old workers in high-level jobs
$w_{\ell,0}^o$	0.4	Initial wages for old workers in low-level jobs

1.1 Further plan

Now that the baseline model is running and produces sensible results, I need to consider how to make the model more realistic, and to consider what data I need to be able to estimate the parameters.

References

2 Appendix

Derivatives for notation: Assume throughout that $0 < \alpha < 1$ and $L_t^y, L_t^o > 0$.

$$\begin{aligned}\frac{\partial Y_t}{\partial L_t^y} &= \alpha(L_t^y)^{\alpha-1}(L_t^o)^{1-\alpha} > 0, \\ \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} &= \alpha(\alpha-1)(L_t^y)^{\alpha-2}(L_t^o)^{1-\alpha} < 0, \\ \frac{\partial Y_t}{\partial L_t^o} &= (1-\alpha)(L_t^y)^\alpha(L_t^o)^{-\alpha} > 0, \\ \frac{\partial^2 Y_t}{\partial L_t^o \partial L_t^o} &= -\alpha(1-\alpha)(L_t^y)^\alpha(L_t^o)^{-\alpha-1} < 0, \\ \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} &= \frac{\partial^2 Y_t}{\partial L_t^o \partial L_t^y} = \alpha(1-\alpha)(L_t^y)^{\alpha-1}(L_t^o)^{-\alpha} > 0.\end{aligned}$$

The first-order conditions for the maximization problem are:

$$\begin{aligned}\frac{\partial \pi_t}{\partial l_{\ell,t}^y} &= A_t \frac{\partial Y_t}{\partial l_{\ell,t}^y} - w_{\ell,t}^y = 0 \\ \Leftrightarrow w_{\ell,t}^y &= A_t \frac{\partial L_t^y}{\partial l_{\ell,t}^y} \frac{\partial Y_t}{\partial L_t^y} \\ \Leftrightarrow w_{\ell,t}^{y*} &= A_t \theta_\ell^y \frac{\partial Y_t}{\partial L_t^y}.\end{aligned}$$

since $\frac{\partial L_t^y}{\partial l_{\ell,t}^y} = \frac{\partial(\theta_h^y l_{h,t}^y + \theta_\ell^y l_{\ell,t}^y)}{\partial l_{\ell,t}^y} = \theta_\ell^y$. The second first order condition is

$$\begin{aligned}\frac{\partial \pi_t}{\partial l_{h,t}^y} &= A_t \frac{\partial Y_t}{\partial l_{h,t}^y} - w_{h,t}^y - cK_t = 0 \\ \Leftrightarrow w_{h,t}^{y*} &= A_t \theta_h^y \frac{\partial Y_t}{\partial L_t^y} - cK_t.\end{aligned}$$

Using $w_{h,t}^y = \mu^y w_{\ell,t}^y$ and isolating for K_t , we find

$$\begin{aligned}A_t \theta_h^y \frac{\partial Y_t}{\partial L_t^y} - cK_t &= \mu^y A_t \theta_\ell^y \frac{\partial Y_t}{\partial L_t^y} \\ \Leftrightarrow \frac{A_t \frac{\partial Y_t}{\partial L_t^y}}{c} (\theta_h^y - \mu^y \theta_\ell^y) &= K_t^*\end{aligned}$$

Average wages: Since $l_t^y = N_t^y$ is fixed at time t and μ^y is constant, we have

$$\begin{aligned}
\frac{\partial \bar{w}_t^y}{\partial l_{h,t-1}^y} &= \frac{\partial \left(w_{\ell,t}^y - \frac{l_{h,t}^y}{l_t^y} w_{\ell,t}^y + \frac{l_{h,t}^y}{l_t^y} \mu^y w_{\ell,t}^y \right)}{\partial l_{h,t-1}^y} \\
&= \frac{\partial \left(\frac{1}{l_t^y} (\mu^y - 1) l_{h,t}^y w_{\ell,t}^y + w_{\ell,t}^y \right)}{\partial l_{h,t-1}^y} \\
&= \frac{1}{l_t^y} (\mu^y - 1) \left(l_{h,t}^y \frac{\partial w_{\ell,t}^y}{\partial l_{h,t-1}^y} + \frac{\partial l_{h,t}^y}{\partial l_{h,t-1}^y} w_{\ell,t}^y \right) + \frac{\partial w_{\ell,t}^y}{\partial l_{h,t-1}^y} \\
&= \frac{1}{l_t^y} (\mu^y - 1) w_{\ell,t}^y \frac{\partial l_{h,t}^y}{\partial l_{h,t-1}^y} + \left[\frac{1}{l_t^y} (\mu^y - 1) l_{h,t}^y + 1 \right] \frac{\partial w_{\ell,t}^y}{\partial l_{h,t-1}^y}
\end{aligned}$$

First, we show the sign of $\frac{\partial l_{h,t}^y}{\partial l_{h,t-1}^y}$, where we use $K_t = l_{h,t}^o + l_{h,t}^y \Leftrightarrow l_{h,t}^y = K_t - l_{h,t}^o \Leftrightarrow l_{h,t}^y = K_t - \rho_h l_{h,t-1}^y$ and it gives us

$$\frac{\partial l_{h,t}^y}{\partial l_{h,t-1}^y} = \frac{\partial K_t}{\partial l_{h,t-1}^y} - \rho_h$$

Now we show

$$\begin{aligned}
\frac{\partial K_t}{\partial l_{h,t-1}^y} &= \frac{\partial \left(\frac{A_t (\theta_h^y - \mu^y \theta_\ell^y)}{c} \frac{\partial Y_t}{\partial L_t^y} \right)}{\partial l_{h,t-1}^y} \\
&= \frac{A_t (\theta_h^y - \mu^y \theta_\ell^y)}{c} \left(\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} \frac{\partial L_t^y}{\partial l_{h,t-1}^y} + \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} \frac{\partial L_t^o}{\partial l_{h,t-1}^y} \right)
\end{aligned}$$

where

$$\begin{aligned}
\frac{\partial L_t^y}{\partial l_{h,t-1}^y} &= \frac{\partial (\theta_h^y l_{h,t}^y + \theta_\ell^y (l_t^y - l_{h,t}^y))}{\partial l_{h,t-1}^y} = \theta_h^y \left(\frac{\partial K_t}{\partial l_{h,t-1}^y} - \rho_h \right) - \theta_\ell^y \left(\frac{\partial K_t}{\partial l_{h,t-1}^y} - \rho_h \right) = (\theta_h^y - \theta_\ell^y) \left(\frac{\partial K_t}{\partial l_{h,t-1}^y} - \rho_h \right) \\
\frac{\partial L_t^o}{\partial l_{h,t-1}^y} &= \frac{\partial (\theta_h^o l_{h,t}^o + \theta_\ell^o l_{\ell,t}^o)}{\partial l_{h,t-1}^y} = \frac{\partial (\theta_h^o \rho_h l_{h,t-1}^y + \theta_\ell^o l_{\ell,t}^o)}{\partial l_{h,t-1}^y} = \theta_h^o \rho_h
\end{aligned}$$

since $\frac{\partial l_{\ell,t-1}^y}{\partial l_{h,t-1}^y} = 0$ by assumption. Substituting this back into the original expression, we get

$$\begin{aligned}
\frac{\partial K_t}{\partial l_{h,t-1}^y} &= \frac{A_t (\theta_h^y - \mu^y \theta_\ell^y)}{c} \left(\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} (\theta_h^y - \theta_\ell^y) \left(\frac{\partial K_t}{\partial l_{h,t-1}^y} - \rho_h \right) + \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} \theta_h^o \rho_h \right) \\
&= \frac{A_t (\theta_h^y - \mu^y \theta_\ell^y)}{c} \left(\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} (\theta_h^y - \theta_\ell^y) \frac{\partial K_t}{\partial l_{h,t-1}^y} + \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} \theta_h^o \rho_h - \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} (\theta_h^y - \theta_\ell^y) \rho_h \right)
\end{aligned}$$

Rearranging terms gives us

$$\begin{aligned}
& \frac{\partial K_t}{\partial l_{h,t-1}^y} - \frac{A_t (\theta_h^y - \mu^y \theta_\ell^y)}{c} \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} (\theta_h^y - \theta_\ell^y) \frac{\partial K_t}{\partial l_{h,t-1}^y} \\
&= \frac{A_t (\theta_h^y - \mu^y \theta_\ell^y)}{c} \left(\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} \theta_h^o - \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} (\theta_h^y - \theta_\ell^y) \right) \rho_h \\
&\Leftrightarrow \frac{\partial K_t}{\partial l_{h,t-1}^y} = \frac{A_t (\theta_h^y - \mu^y \theta_\ell^y)}{c - A_t (\theta_h^y - \mu^y \theta_\ell^y) \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} (\theta_h^y - \theta_\ell^y)} \left(\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} \theta_h^o - \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} (\theta_h^y - \theta_\ell^y) \right) \rho_h
\end{aligned}$$

The above term is less than ρ_h when $c > \bar{c} = A_t (\theta_h^y - \mu^y \theta_\ell^y) \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} \theta_h^o$. Therefore,

$$\frac{\partial l_{h,t}^y}{\partial l_{h,t-1}^y} = \frac{\partial K_t}{\partial l_{h,t-1}^y} - \rho_h < 0$$

and the first term in average wage equation is negative,

$$\frac{1}{l_t^y} (\mu^y - 1) w_{\ell,t}^y \frac{\partial l_{h,t}^y}{\partial l_{h,t-1}^y} < 0$$

Finally, for young workers in low level positions, we have

$$\begin{aligned}
\frac{\partial w_{\ell,t}^{y*}}{\partial l_{h,t-1}^y} &= \frac{\partial \left(A_t \theta_\ell^y \frac{\partial Y_t}{\partial L_t^y} \right)}{\partial l_{h,t-1}^y} = A_t \theta_\ell^y \left(\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} \frac{\partial L_t^y}{\partial l_{h,t-1}^y} + \frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} \frac{\partial L_t^o}{\partial l_{h,t-1}^y} \right) \\
&= A_t \theta_\ell^y \left(\underbrace{\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^y} (\theta_h^y - \theta_\ell^y)}_{<0} \underbrace{\left(\frac{\partial K_t}{\partial l_{h,t-1}^y} - \rho_h \right)}_{<0} + \underbrace{\frac{\partial^2 Y_t}{\partial L_t^y \partial L_t^o} \theta_h^o \rho_h}_{>0} \right)
\end{aligned}$$

under the above condition for c . Hence, we have shown what Bianchi and Paradisi have shown.